

0 5 . 1

Describe the role of glucagon in gluconeogenesis.

Do **not** include in your answer details on the second messenger model of glucagon action.

[2 marks]

0 5 . 2

The gene that codes for glucagon is 9.531 kilobases in length. The DNA helix makes one complete turn every 10 base pairs. Every complete turn is 3.4 nm in length.

Use this information to calculate the length in micrometres (μm) of the gene for glucagon. Give your answer to 3 significant figures.

[2 marks]

Answer = _____ μm

Metformin is a drug commonly used to treat type II diabetes. Metformin's ability to lower the blood glucose concentration involves a number of mechanisms including:

- increasing a cell's sensitivity to insulin
- inhibiting adenylate cyclase.

0 5 . 3

Explain how increasing a cell's sensitivity to insulin will lower the blood glucose concentration.

[2 marks]

0 5 . 4

Explain how inhibiting adenylate cyclase may help to lower the blood glucose concentration.

[3 marks]
